

ZNOTES.ORG

UPDATED TO 2023-25 SYLLABUS

CAIE IGCSE CHEMISTRY

SUMMARIZED NOTES ON THE THEORY SYLLABUS

1. States of Matter

1.1. Solids, liquids and gases

- States of matter are the different forms in which matter can exist
- The three states of matter are: Solid, liquid and gases

Characteristics of Solids, liquids and gases -->

	SOLIDS	LIQUIDS	GASES
SHAPE	have a definite shape	take the shape of their container	take the shape of their container
DENSITY	high	moderate to high	low
VOLUME	have a fixed volume	have a fixed volume	don't have a fixed volume - expand to fill the container
FLUIDITY	non-fluid in nature	fluid in nature	fluid in nature
INTER-PARTICULAR SPACES	closely-packed together, barely any inter-molecular spaces	closely-packed together, barely any inter-molecular	presence of large, inter-molecular spaces
PARTICULAR MOVEMENT	particles are immobile, however, they can vibrate about their fixed positions	particles are able to move past one another	particles are mobile and move randomly
PARTICULAR ARRANGEMENT	particles are arranged regularly in a lattice	particles are arranged irregularly	particles are arranged irregularly

1.2. Boiling and Evaporation

Boiling and Evaporation are (endothermic processes):

- Boiling and evaporation are processes which involve the conversion of a liquid into a gas.
- Both processes allow **molecules to move further apart from each other.**

Boiling	Evaporation
boiling happens at a set temperature called the boiling point	evaporation occurs at any temperature below the boiling point and above the freezing point (liquid)

Boiling	Evaporation
boiling occurs throughout the liquid	evaporation only occurs at the surface
boiling is a relatively faster process	evaporation is a slow process

Condensation

Condensation is the process in which a gas converts into a liquid. It happens at the same temperature as the boiling point

- As temperature decreases, the **energy of particles will decrease**, making it move more slowly.
- Condensation and Freezing are both **energy given out reaction.**

Freezing and melting (endothermic processes)

- Melting is the process in which a solid converts to a liquid. It happens at a set temperature called the melting point
- Freezing is the process in which a liquid converts to a solid. It happens at the same temperature as the melting point

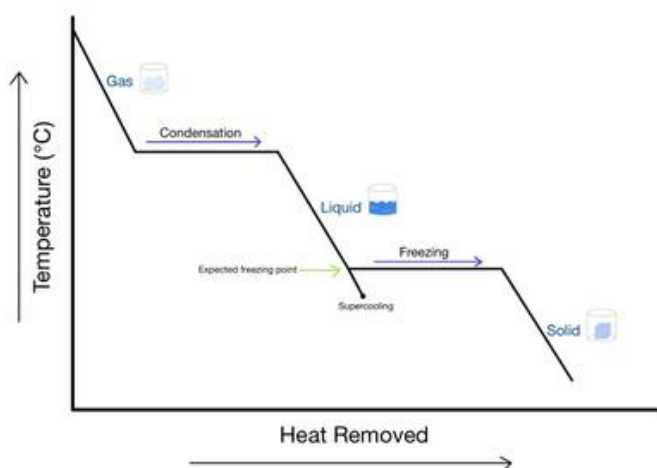
1.3. Cooling and Heating Curves (Extended)

Cooling curves

- On cooling, the particles of a gas move slower and slower and the gas contracts
- The particles are now closer together and inter-molecular bonds start to form between them once the condensation point is reached
- The temperature of the substance stops falling; the energy released by bond formation cancels out the energy lost due to cooling
- Once all the gas has turned into liquid, the temperature starts to fall again and the liquid begins to contract until the freezing point is reached
- At the freezing point, inter-molecular bonds between the liquid molecules start to develop to form a solid
- At the freezing point, the temperature remains constant until all of the liquid has solidified since, the energy released due to bond formation cancels out the energy lost due to cooling

The following curve is obtained if this phenomenon is plotted - the cooling curve -->

Phase Change Diagram

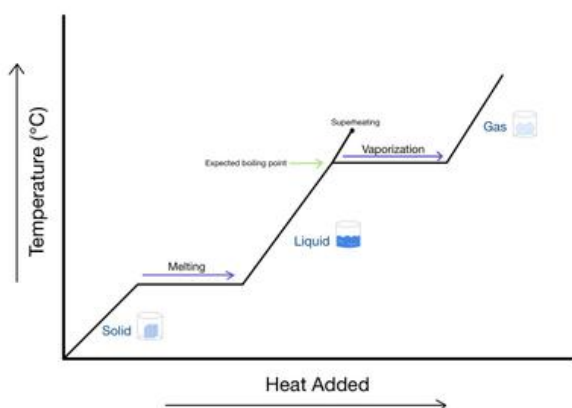


Heating Curves

1. On heating, the particles of a solid start to vibrate faster and faster in their mean positions and the solid begins to expand
2. Once the melting point is reached, the inter-molecular bonds between the particles begin to breakdown
3. The temperature of the substance at this point remains constant until all the solid has turned into a liquid because the energy received by the system is cancelled out by the energy used to break intermolecular bonds
4. Once all the solid has turned to liquid, the temperature starts to rise again and the liquid begins to expand until the boiling point is reached
5. At the boiling point, the intermolecular bonds between the liquid molecules start to break down to form gas
6. At the boiling point, the temperature of the substance remains constant until all of the liquid has vaporized since, the energy absorbed by the substance is cancelled by the energy used for breakdown of intermolecular bonds

The following curve is obtained if this phenomenon is plotted - the heating curve →

Phase Change Diagram



1.4. Effects of Temperature and Pressure on Gas

Gases are compressible in nature. By changing the pressure acting on them, their volume may be influenced.

1. An increase in external pressure produces a contraction (decrease) in volume. The gas is said to be compressed
2. A fall in external pressure produces an expansion (increase) in volume. The gas is said to be decompressed

The volume of gases may also be influenced by temperature. The temperature of a gas affects its internal pressure and thereby its volume.

1. When temperature increases, the gas molecules have increased kinetic energy and hit the walls of their container more often and with greater force. This causes an increase in internal pressure and an increase in volume.
2. When temperature decrease, the gas molecules have decreased kinetic energy and hit the walls of their container less often and with attenuated force. This causes a decrease in internal pressure and a decrease in volume.

In conclusion →

1. Temperature and volume of gases are **directly proportional**
2. External pressure and volume of gases are **inversely proportional**

1.5. Diffusion

Diffusion, the net movement of particles from a **region of higher concentration to a region of lower concentration**, as a result of their random movement until equilibrium is reached. The rate of diffusion is most rapid in **gases > liquids > solids**

1.6. Effect of Relative Molecular Mass in Diffusion (Extended)

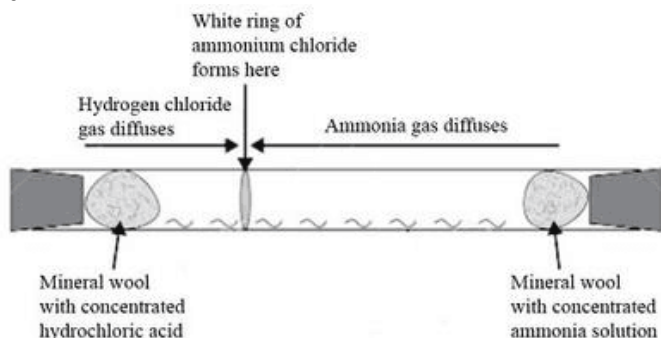
- The rate at which gases diffuse is different and is dependent on the mass of the gas molecules.
- At the same temperature, molecules that have a lower mass move, on average, faster than those with a higher mass.

The HCl and NH₃ experiment

The experiment involved placing a cotton wool plug soaked in hydrochloric acid (HCl) at one end of a glass tube and a cotton wool plug soaked in ammonia (NH₃) at the other end.

The tube was left undisturbed for a period of time, during which the gases diffused towards each other. The rate of diffusion was measured by observing the distance travelled by each gas after a fixed time. It was found that ammonia, with a lower molecular mass, diffused faster than hydrochloric acid, which has a higher molecular mass.

Conclusion: Ammonia has a **smaller Mr** than hydrochloric acid so ammonia has a **greater rate of diffusion**. Therefore, the white ring of ammonium chloride **forms near the hydrochloric acid**.



2. Atoms, Elements and Compounds

Atom: The smallest particle of matter made of protons, neutrons and electrons

2.1. Elements, Compounds and Mixtures

- Elements → Group of atoms that share the **SAME number of protons**. It cannot be broken down into simpler substances by chemical methods
- Compounds → Two or more elements **chemically bonded together**
- Mixtures → Two or more elements **not chemically bonded together**

2.2. Atomic Structure

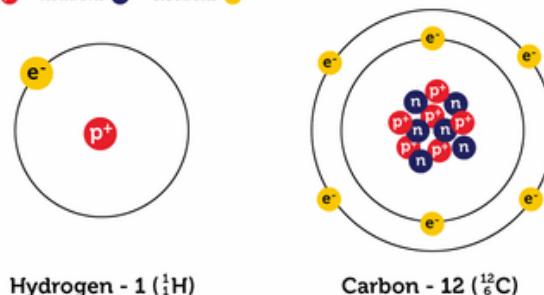
- An atom is made up of three subatomic particles: protons, neutrons and electrons
- The atom is mostly space with a positively charged nucleus consisting of protons and neutrons in the centre and electrons in the empty space around the nucleus (held together by the electrostatic force of attraction between them and the positively charged nucleus)
- The characteristics of neutrons, protons and electrons are as follows →

Subatomic particle	Relative Mass	Relative Charge
proton	1	+1
neutron	1	0

Subatomic particle	Relative Mass	Relative Charge
electron	1/1840	-1

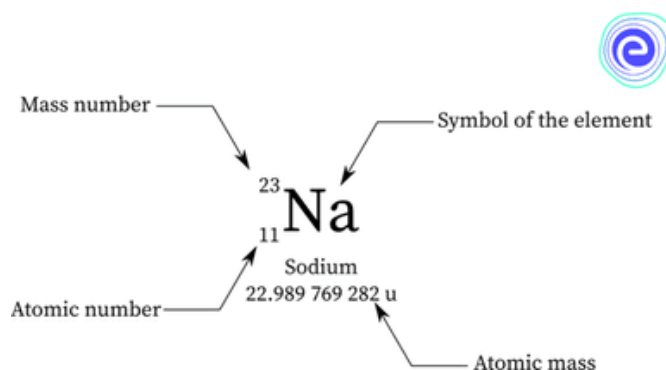
- Since electrons and protons have opposing and equal charges, **the atom's overall charge is neutral**.
- Neutrons have the purpose of holding the nucleus together. The larger the nucleus gets, more are the neutrons required to hold the nucleus together

protons p^+ neutrons n electrons e^-



2.3. Proton Number and Nucleon Number

- The **proton number (atomic number)** is the number of protons in the nucleus of an atom.
- It is unique to each element. It is denoted with the letter "Z". For an atom which is neutral in nature, the number of protons and the number of electrons are equal therefore the proton number (Z) also corresponds to the number of electrons
- The **nucleon number (mass number)** is the total number of protons and neutrons in the nucleus of an atom. It is denoted with the symbol "A".
- According to scientific convention, the mass number and the atomic number of an element is combined in the symbol itself in the following format →



2.4. Electronic Configurations in the Periodic Table

Atoms have electrons that orbit around a central nucleus, and these orbits are referred to as electron shells.

1. The energy levels of the shells increase as their distance from the nucleus increases.
2. The first shell has a maximum capacity of **2 electrons**, while each subsequent shell can hold up to **8 electrons**.

For this syllabus, we only need to know the **general full electronic configuration as (2.8.8)**

- (a) Group VIII noble gases have a **full outer shell**
- (b) the **number of outer shell electrons** is equal to the **group number in Groups I to VII**
- (c) the number of occupied electron shells is **equal to the period number**

2.5. Calculating Relative Atomic Masses (Extended)

- Most elements exist naturally as a mixture of their isotopes. Using the data on the abundance of these naturally occurring isotopes, we can calculate the mass relative atomic mass of the element.
- NOTE → for all purposes, the mass numbers of elements have been rounded off to the nearest whole number however, only Chlorine is used with its actual mass number of 35.5. This is for the sake of simplicity in calculation
- An example for calculating relative mass and abundances →

Q. Iridium has two isotopes. These isotopes are Iridium - 191 and Iridium - 193. A natural sample consists of 37.3% of Iridium - 191. Calculate the relative atomic mass (A_r) of the natural sample of Iridium

A. →

Step 1. Identify the percentage of Iridium - 193

→ if the sample consists of 37.3% of Iridium - 191, it must consist of 100% - 37.3% of Iridium - 193

→ 100 - 37.3 = 62.7%

→ the sample consists of 62.7% of Iridium - 193

Step 2. Consider a sample of 100 atoms of Iridium. In that sample, 37.3 of atoms should have a mass of 191 and 62.7 atoms should have the mass of 193

THEN

$$\text{average mass} = \frac{(37.3 \times 191) + (62.7 \times 193)}{100} = 192.2 \text{ (4 significant figures)}$$

Answer: 192.4

2.6. Isotopes

Isotopes → different atoms of the same element that have the same number of protons but different numbers of neutrons.

- The isotopes of an element have the **same chemical properties** because **they contain the same number of outer shell electrons** and therefore have the **same electronic configuration**.
- Radioisotopes → certain isotopes such as Carbon - 14 and Tritium have a nucleus so heavy that they are radioactive in nature. Their nucleus is **unstable and breaks up spontaneously**

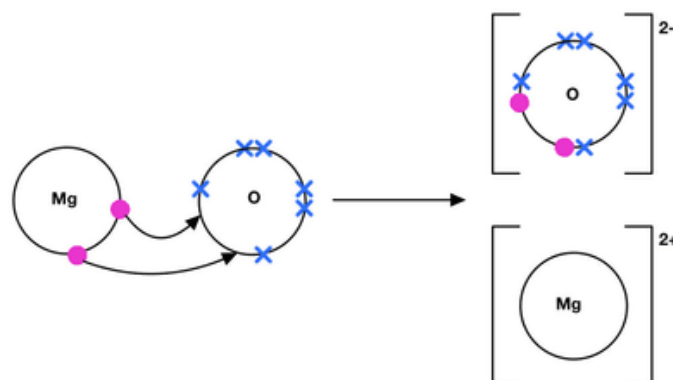
2.7. Ions and ionic bonds

Cations - Positive Ions

Anions - Negative Ions

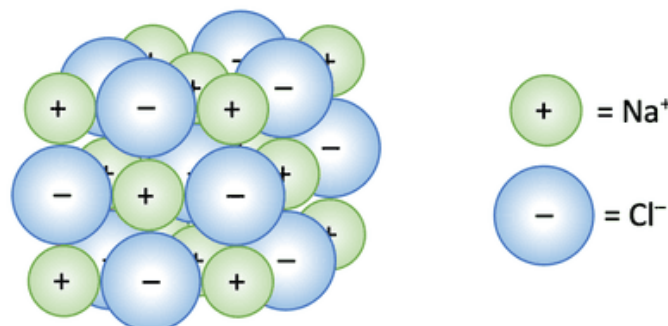
Ionic Bonds - a strong electrostatic attraction between oppositely charged ions (metals + non metals)

Dot and Cross Diagram



Ionic Compounds

Typical ionic compounds are usually a **giant lattice structure** with **regular arrangement of alternating positive and negative ions**.



Properties of Ionic Compound

1. High Boiling and Melting Point - A lot of energy needed to overcome the strong electrostatic forces between oppositely charged ions
2. Good Electrical Conductivity when **molten or aqueous**:
 - Ions are free to move in aqueous/molten state as the **charges can flow**.
 - Ions are not free to move when solid state as the **charges cannot flow**.
3. Ionic Compounds are **soluble in water** - The positive and negative ions are **attracted to water**
4. **Solid in Room temperature** - Regular arrangement of ions in a lattice with ions of opposite charges next to each other

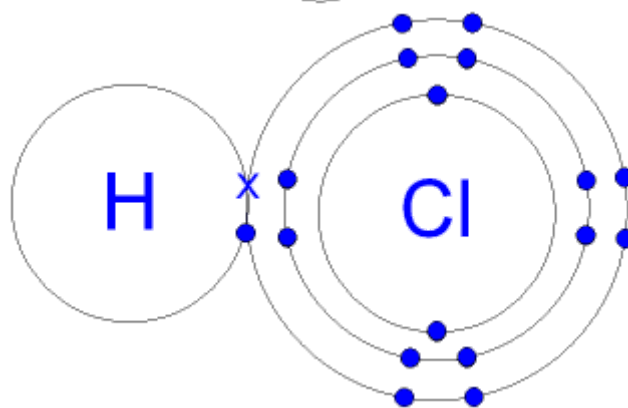
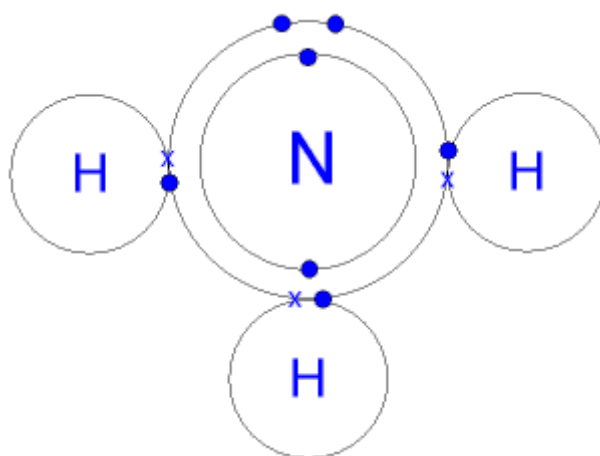
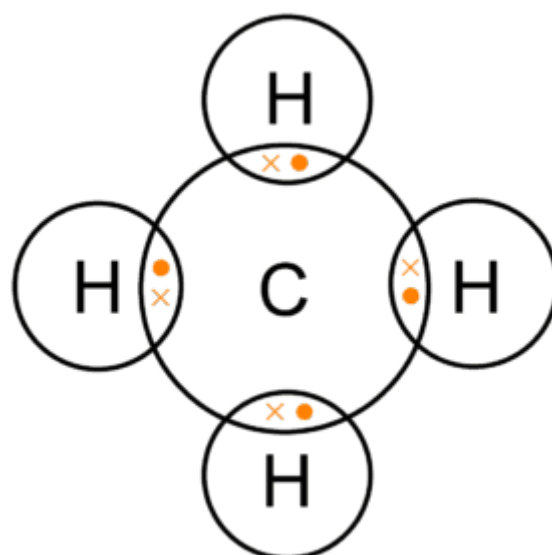
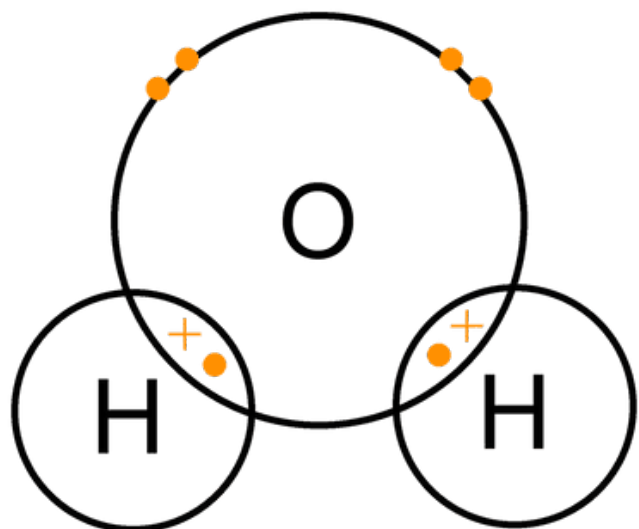
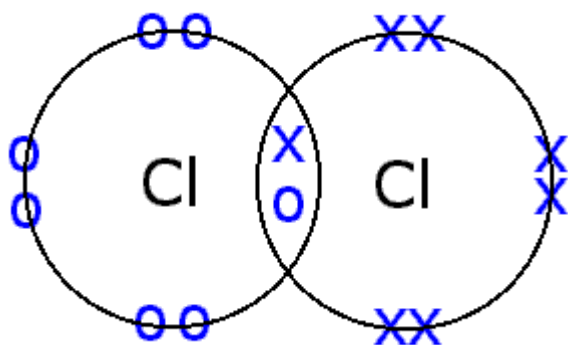
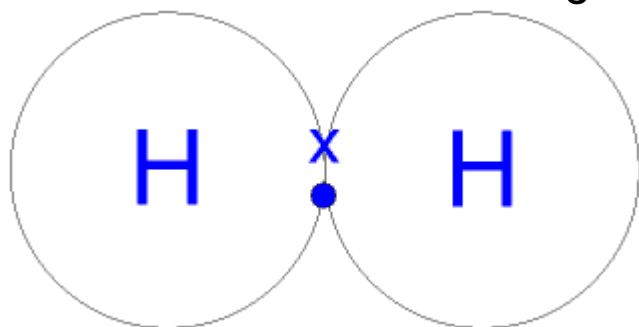
Other Properties of Ionic Compound

- Brittle
- Low Volatility

2.8. Simple Molecules and Covalent Bonds

Covalent Bonding are pairs of electrons is shared between two atoms leading to noble gas electronic configuration (2.8.8)

Covalent Bonds Dot and Cross Diagram



Different types of Covalent Bonds

1. Single Bonds - E.g: Chlorine
2. Double Bonds - E.g: Carbon Dioxide
3. Triple Bonds - E.g: Nitrogen

Properties of Covalent Bonding

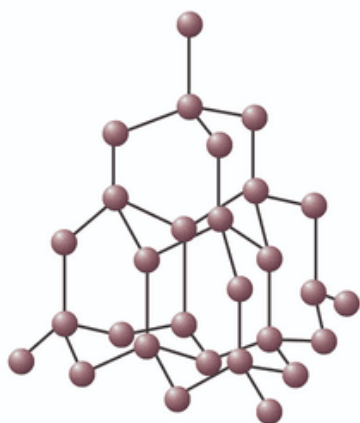
1. The intermolecular forces in covalent compounds are **weak**, but they have **strong covalent bonds**.
2. Covalent Compounds have low melting and boiling point - They require ****less energy** to break the weak intermolecular forces (****same term as attractive forces**).

3. Poor Electrical Conductivity - **No free electrons or ions present** to carry an electrical current

2.9. Giant Covalent Structures

Substances with giant covalent structures (macromolecular structures) are **solids with very high melting points**. All the atoms are linked by **strong covalent bonds** and **made of Pure Carbon**.

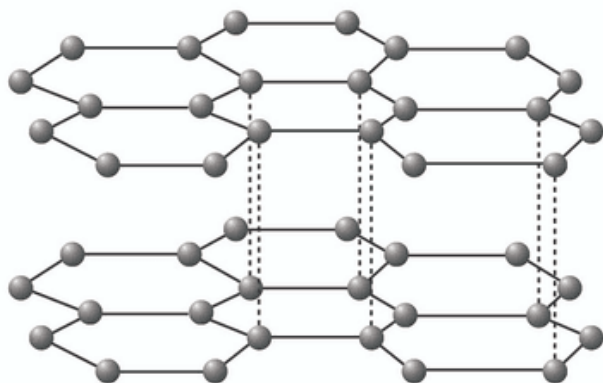
Diamond



Properties of Diamond

1. Each carbon atoms are joined with **4 other carbon atoms**
2. **High Melting and Boiling Points** - Strong Covalent Bonds
3. **No Delocalised/Free Moving Electrons**
4. Cannot be scratched easily
5. Transparent colour
6. Cannot conduct electricity **due to no free moving electrons**
7. **Hard** in structure
8. **Giant Lattice Arrangement**
9. Uses are for **cutting tools**

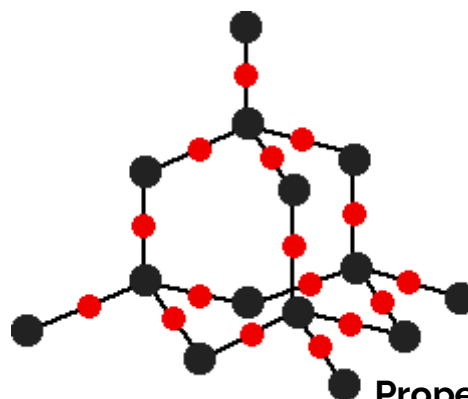
Graphite



Properties of Graphite

1. Each carbon atoms are joined with **3 other carbon atoms**
2. **High Melting and Boiling Points** - Strong covalent bonds within the layers (but the layers are attracted to each other by weak intermolecular forces)
3. Contains **Delocalised/Free Moving Electrons**
4. Can be scratched easily
5. Opaque/Black
6. Can conduct electricity **due to free moving electrons**
7. Soft - **Layers can slide easily**
8. **Layers of hexagonal rings** held by weak intermolecular forces
9. Uses are for **lubricant and electrode** in Electrolysis

Silicon (IV) Oxide (SiO₂)

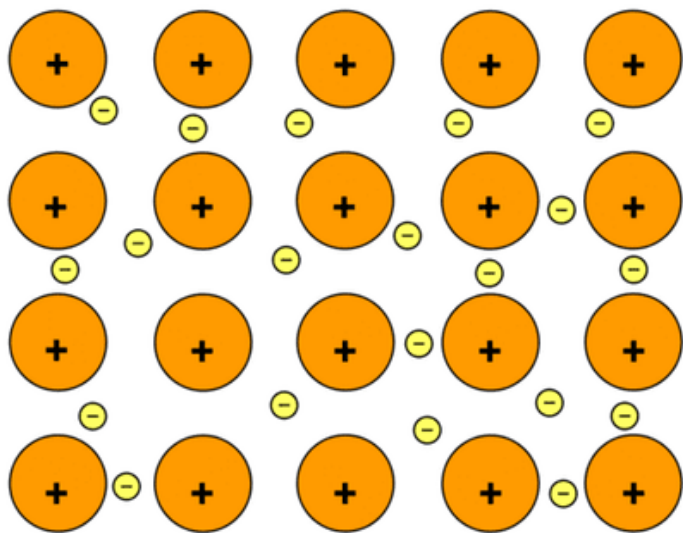


Properties of Silicon (IV) Oxide

1. Structure of Silicon (IV) Oxide is **similar/resemblance to diamond**
 2. **Hard Structure**
 3. **High melting and boiling point** - More energy to overcome
 4. **Rigid Tetrahedral Structure**
 5. Does **not** conduct electricity
- Each **Silicon atom** is covalently bonded with **4 Oxygen Atoms**
 - Each **Oxygen atom** is covalently bonded with **2 Silicon Atoms**

2.10. Metallic Bonding

Metallic Bonding is the **electrostatic attraction** between the **positive ions** in a **giant metallic lattice** and a **"sea" of delocalised electrons**. (*Remember this to perfection)



Structure and Bonding

1. Metallic Bonds have good electrical conductivity - **Delocalised electrons can move through the structures and carry current.**
2. High Melting and Boiling Point - **More energy to overcome strong forces of attraction between positive metal ions and the sea of delocalised electrons & Vibrate/Transfer Heat**
3. Malleability - Can be **hammered into shapes** as **layers can slide over each other**
4. Ductility - Can be **drawn into thin wires**

3. Stoichiometry

3.1. Formulae

- Charges of each elemental group

Group	Charges
I	1
II	2
III	3
IV	+/- 4
V	-3
VI	-2
VII	-1
VIII	0

- Charges of common ions

Name	Formula	Charges
Iron (II)	Fe^{2+}	+2
Copper (II)	Cu^{2+}	+2
Ammonium	NH_4^+	+1
Nitrate	NO_3^-	-1
Hydroxide	OH^-	-1

Name	Formula	Charges
Carbonate	CO_3^{2-}	-2
Sulfate	SO_4^{2-}	-2
Phosphate	PO_4^{3-}	-3

3.2. State Symbols & Word equations

- Balancing equations: A chemical equation is balanced when there are equal number of atoms and charges on both sides of the equation
- State symbols:
 - (s) = solid
 - (l) = liquid
 - (g) = gas
 - (aq) = aqueous solution
- Names of compounds
 - Compound ending with **-ide** only contain two different elements
 - Compound ending with **-ate** contain oxygen

3.3. Relative Masses of Atoms and Molecules

- Relative atomic mass (A_r):** the average mass of the isotopes of an element compared to 1/12th of the mass of an atom of ^{12}C
- Relative molecular mass (M_r):** sum of relative atomic masses of all the atoms in one molecule of the compound

3.4. The mole and the Avogadro Constant

A **mole** of a substance is the amount that contains the same number of units as the number of carbon atoms in 12 grams of carbon-12

- A mole is the A_r or M_r expressed in grams e.g. 1 mole of Carbon-12 is equal to 12 grams.
- It is equal to 6.02×10^{23} particles, this number is called Avogadro's constant.
- 1 mole also occupies a volume of 24dm^3 at room temperature and pressure

3.5. Number of Moles

$$\text{Amount of Substance} = \frac{\text{mass}}{\text{molar mass}}$$

- Amount of Substance (mol)
- Mass (g)
- Molar Mass (M_r) in (g/mol)

3.6. Number of Moles in Aqueous Solution

Moles = concentration x volume

- Moles (mol)
- Concentration (mol/dm³)
- Volume (24 dm³)

3.7. Moles in Gases

Volume = No. of Moles × 24dm³ at r.t.p (room temperature and pressure)

- 1 dm³ = 1000cm³

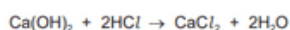
3.8. Concentration

$$\text{Concentration} = \frac{\text{no. of moles}}{\text{volume}}$$

- Moles per dm³
 - 1mol/dm³
- Grams per dm³, g/dm³

Concentration can be measured and converted into g/dm³ or mol/dm³, by multiplying the molar mass of the compound.

3.9. Moles Calculation in Acid-Base Titrations



20.0cm³ of 0.0500mol/dm³ HCl reacts with the 25.0cm³ of Ca(OH)₂.

Determine the concentration of Ca(OH)₂ in g/dm³. Use the following steps.

- Calculate the number of moles in 20.0cm³ of 0.0500mol/dm³ HCl.

..... mol

We will use the formula:

$$\text{Concentration} \times \text{Volume (cm}^3\text{)} / 1000$$

$$20 \times 0.0500 \text{ divide } 1000 = 0.001$$

3.10. Empirical Formulae

- This is the simplest ratio of the atoms in a compound
- For example:
 - Molecular formula of ethane = C₂H₆
 - Empirical formula of ethane = CH₃
- To find out the empirical formula you:
 - Make the percent ratio into the simplest whole number ratio (NOTE: if given %s, use them as grams)
 - Divide the coefficients of each element symbol by the lowest coefficient

3.11. Molecular Formulae

- It shows the actual number of atoms in one molecule of a substance.

3.12. Percentage Purity & Yield

- Percentage purity = $\frac{\text{mass of product (pure)}}{\text{mass of compound (impure)}} \times 100$
- Percentage yield = $\frac{\text{actual mass obtained}}{\text{calculated/theoretical mass}} \times 100$

3.13. Percentage Composition by Mass

This is the Percentage Composition by Mass Formula
(mass of element/molecular mass) x 100

Example Question (0620/42/F/M/23)

Calculate the percentage composition by mass of nitrogen in (NH₄)₂SO₄.

percentage of nitrogen = % [2]

Step 1: Calculate the molar mass of (NH₄)₂SO₄

Step 2: Find the mass of Nitrogen

Step 3: Use the formula and multiply with 100.

Step 4: That will be the percentage of the question given.

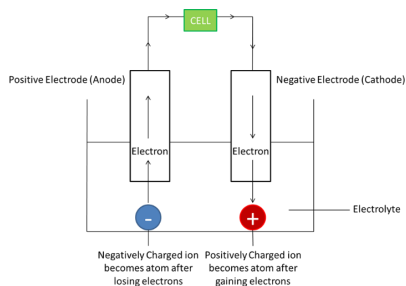
4. Electrochemistry

It is the decomposition of an ionic compound, when molten or aqueous solution- by passing of an electric current

- This is possible due to the presence of **mobile electrons/free moving electrons**
- Electrolyte is a **molten or aqueous substance that undergoes electrolysis**

Components of Electrolysis	Definition
Electrodes	Metal or graphite rods that aid the flow of electricity in and out of the electrolyte 1. Anode: Positive electrode 2. Cathode: Negative Electrode (PANIC - Positive is Anode, Negative is Cathode)
Anion	Negatively charged ion that moves to anode
Cation	Positively charged ion that moves to cathode

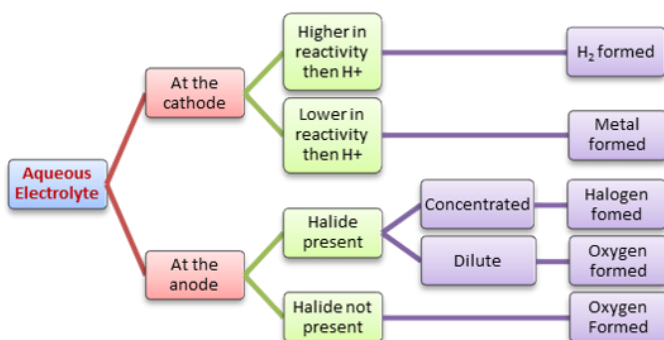
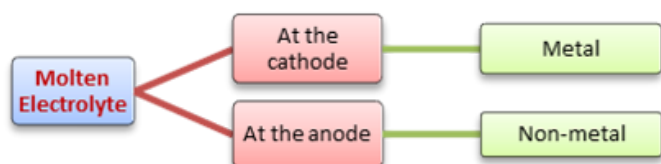
- Note: **Reactive** electrodes take part in the reaction while **inert** electrodes (Graphite, Carbon) do not react with the cations or anions.



4.2. Electrolysis Continued

- Reduction of positive cations happens at the cathode
 - Oxidation of negative anions happens at the anode
- For example (Ionic Half Equations)

- At the anode: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
- At the cathode: $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$



Useful Acronyms to remember

- REDCATANOX (Reduction is Cathode, Anode is Oxidation)
- OILRIG (Oxidation is loss, reduction is gain)
- PANIC (Positive is Anode, Negative is cathode)
- CMAN (Cathode discharge Metals, Anode Discharge Non-Metals)

4.3. Observations in Electrolysis

Electrolyte	At cathode	At anode
Molten lead (II) bromide	Lead	Bromine
Concentrated hydrochloric acid	Hydrogen	Chlorine
Concentrated aqueous sodium chloride (Brine)	Hydrogen	Chlorine
Dilute sulfuric acid	Hydrogen	Oxygen
Aqueous Copper (II) Sulfate with Graphite Electrodes	Copper	Oxygen \n

Electrolyte	At cathode	At anode
Aqueous Copper (II) Sulfate with Carbon Electrodes	Copper	Copper \n

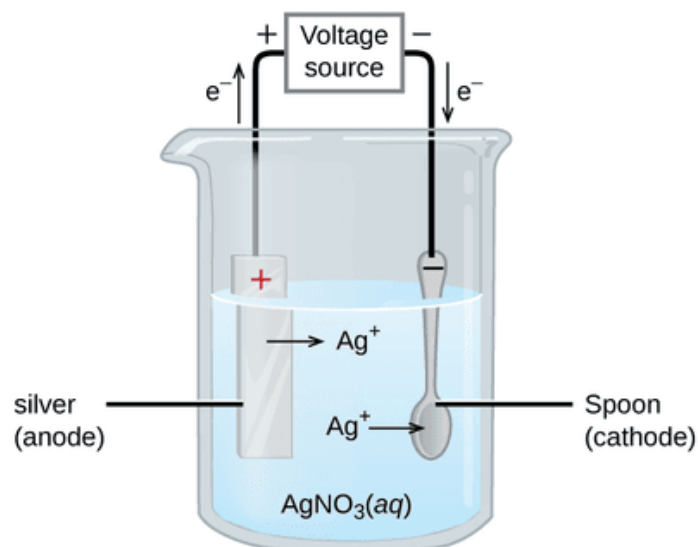
*Blue copper (II) sulfate doesn't change as the concentration of Cu^{2+} ions remains the same

Inert (Unreactive electrodes) are **Platinum, Graphite or Carbon Electrodes** - So it **doesn't react with the ions during electrolysis**.

4.4. Electroplating

The process of coating the surface of a metal (more reactive) with another metal (less reactive) using electrolysis

- Components:
 - Anode: pure metal being used to electroplate the object
 - Cathode: object being electroplated
 - Electrolyte: aqueous solution of the soluble salt of pure metal (same as anode)
- Used to:
 - Prevent corrosion
 - Enhance appearance



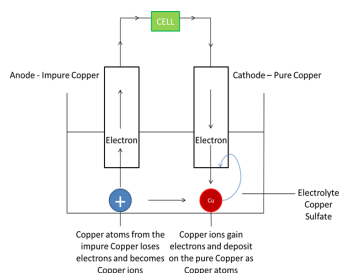
4.5. Refining Metals

- Cathode: thin strip of pure metal
- Anode: impure metal
- Electrolyte: Aqueous Salt Solution of metal

Example:

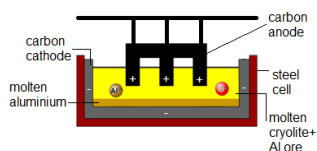
- The refining of copper:** Impure copper as the anode and pure copper as the cathode; the aqueous copper (II) sulfate helps the copper ions move from the anode to the cathode. Here the ions gain electrons and become copper atoms, making the pure copper cathode thicker.

1. Reaction at Anode: $\text{Cu} \rightarrow 2\text{e}^- + \text{Cu}^{2+}$ (mass decreases)
2. Reaction at Cathode: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ (mass increases)



4.6. Extraction of Aluminum

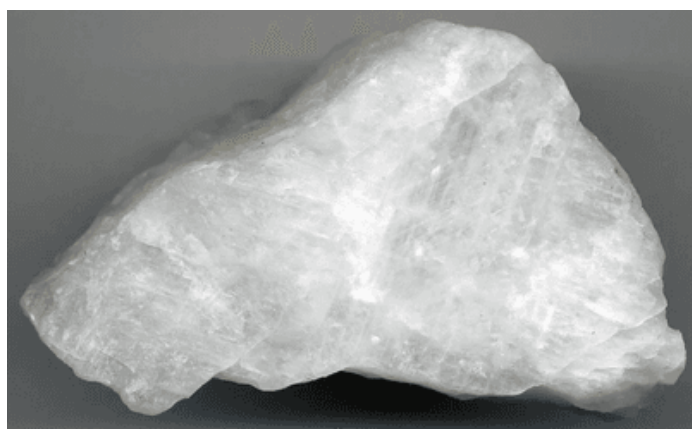
- The main ore of aluminum oxide is bauxite
- Aluminum (III) oxide (alumina) is dissolved in molten cryolite (Na_3AlF_6) – this mixture has a lower m.p. (industrially preferred)



During electrolysis aluminum ($\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$) is produced at the carbon cathode and oxygen ($2\text{O}^{2-} + 4\text{e}^- \rightarrow \text{O}_2$) at the carbon anode.

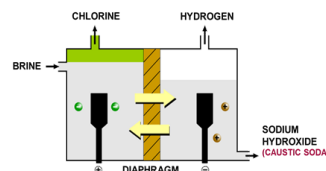
Cryolite

- High temperatures are needed to melt it is expensive. Therefore, the use of molten cryolite **lowers the melting point/operating temperature and increases conductivity**.
- Due to the high temperature, the oxygen reacts with the carbon in the graphite anode to form CO_2 and so **anode has to be periodically replaced**.



4.7. Electrolysis of Brine

- Brine is concentrated aqueous NaCl solution
- Ions present: Na^+ , H^+ , Cl^- and OH^-



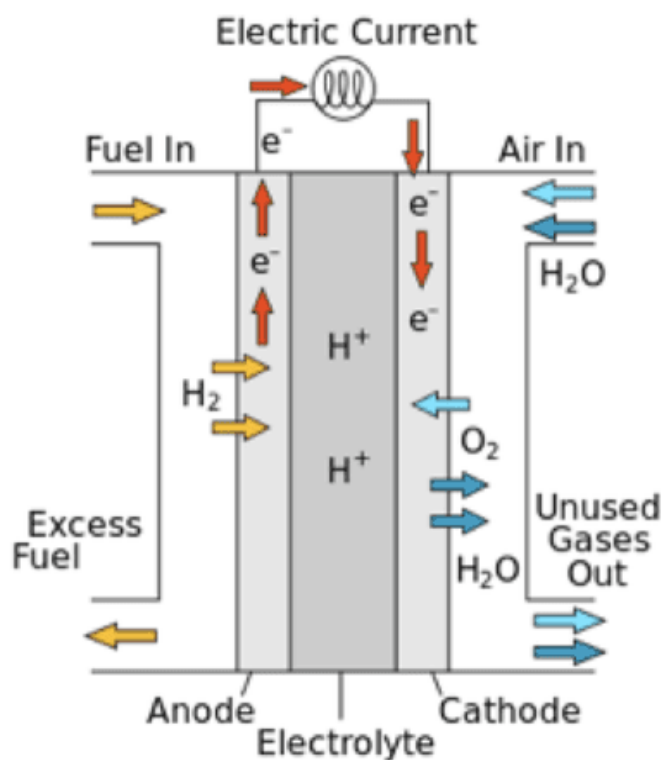
At the anode	At the cathode
Made of titanium	Made of steel
Cl^- ions; Chlorine gas	Hydrogen cations reduced to H_2 molecules

Unreacted ions (Na^+ , H^+ and OH^-) move through porous membrane due to difference in liquid pressure

Left Na^+ and OH^- which form aqueous sodium hydroxide

4.8. Hydrogen-oxygen Fuel Cells

It is the process of using **hydrogen and oxygen as the main reactants** to produce electricity where its only product released is **water**.



Main Chemical Equation: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

Advantages for motor vehicle	Disadvantage for motor vehicle
Renewable source	Large fuel tank required
Lesser flammability from petrol	Currently expensive
Emission Free (No carbon pollutions)	Lesser Hydrogen Filling stations
Non-toxic	

5. Chemical Energetics

5.1. Exothermic & Endothermic Reactions

Chemical reactions involve a transfer of energy between the system (the chemical reaction) and its surroundings.

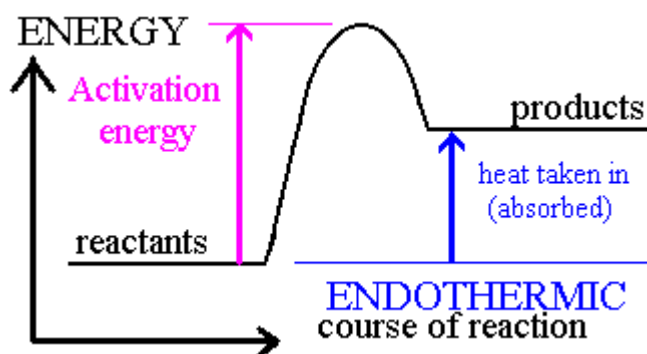
Exothermic reaction	Endothermic reaction
Heat energy is released into the surroundings	Heat energy is absorbed from the surroundings
Bond making reactions	Bond breaking reactions
Surrounding temperature increases	Surrounding temperature decreases

5.2. Energy Level Diagrams

- Energy level diagrams are a representation of chemical reactions that include the relative energies of the reactant and product.
- The energy change (ΔH) of a reaction is represented through the difference in height between the reactant and its product.
- Activation energy (E_a) is the minimum energy required for the reaction to take place

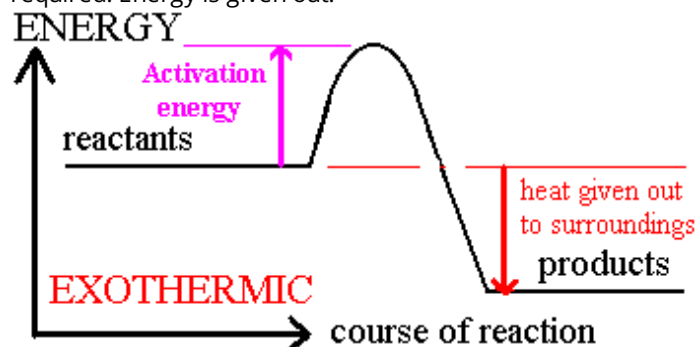
Endothermic energy level diagram:

Energy is gained by the system; higher activation energy required. Energy is taken in.



Exothermic energy level diagram:

Energy is lost by the system; lower activation energy required. Energy is given out.



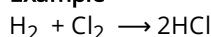
5.3. Bond Energy

- This is the amount of energy required or released when a bond is formed or broken respectively. The unit measure of this energy is kJ/mol.
- The energy change in a reaction is calculated using the following formula:

$$\Delta H = \text{Bond Breaking} - \text{Bond Forming}$$

- If overall heat energy value is negative, reaction is exothermic
- If overall heat energy value is positive, reaction is endothermic

Example



Bond	Bond energy (kJ/mol)
H - H	436
Cl - Cl	243
H - Cl	432

Bond breaking $\rightarrow 436 + 243 = 679$ kJ/mol

Bond forming $\rightarrow 2(432) = 864$ kJ/mol

Thus,

$$\Delta H \rightarrow 679 - 864 = -185 \text{ kJ/mol}$$

The reaction is exothermic in nature

6. Chemical Reactions

Physical change	Chemical change
Reaction is easily reversible	Reaction is harder to reverse
Product has no new chemical properties	Chemical product has different properties
Ex. dissolving solute in a solvent	Energy change [exothermic/endothermic]

6.2. Rates of Reaction

Collision Theory

Successful collisions have enough activation energy at moment of impact to break pre-existing bonds and form new bonds

Rates of reaction

- The measure of the speed of the collision
- Calculated by concentration of reactant used up or product produced per unit time
- Unit = (mol/dm³)/s

6.3. Concentration

Increasing concentration of reactants increases rate of reaction

- Higher concentration reactants contain more particles per unit volume; this increases successful collision rate leading to an increased rate of reaction.
- When the concentration varies or changes in the rate of reaction graph, the ~~collision energy will remain the same~~; only collision rate will increase (and the **activation energy do not change**).



Less particles, less frequent and successful collision



More particles, more frequent and successful collision

6.4. Temperature

Increasing temperature increases the rate of reaction

- Increased temperatures leads to increased average kinetic energy of particles. Particle movement produces energy greater than/equal to activation energy; increased successful collision rate leads to an increased rate of reaction.



Particles have less energy, less frequent and successful collision



Particles have high energy, more frequent and successful collision

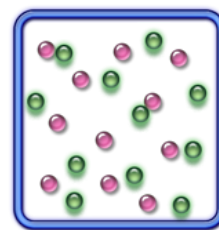
6.5. Surface area of solids

Decreasing the particle size increases the rate of reaction

- Decreasing particle size increases surface area; more reactant particles exposed to collide so the successful collision rate increases. This results in an increased rate of reaction



Less particles, less frequent and successful collision



More particles, more frequent and successful collision

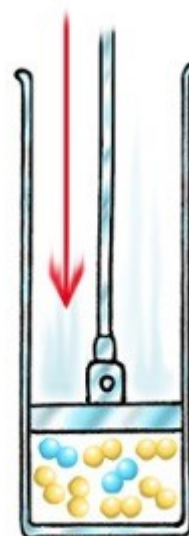
Explosive combustion

- Fine particles are combustible in air due to larger surface area
- The rate of reaction is high making them explosive
- Examples: methane gas in coal mines and flour milling

6.6. Pressure

Increasing the pressure in a gaseous system increases the rate of reaction

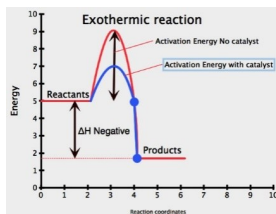
- The distance between particles is reduced under pressure
- There are more particles per unit volume; the successful collision rate increases, resulting in an increased rate of reaction.



6.7. Catalyst

A catalyst is a substance which **increases the rate of reaction by lowering the activation energy** and is left unchanged at the end of the reaction.

- More particles will have an energy greater than or equal to the activation energy, therefore successful collision rate increases resulting in increased rate of reaction
- For gaseous reactants, if catalyst is solid metal, the catalyst provides a surface for reaction to take place on
- The larger the surface area of the metal catalyst, the larger the area for reaction to take place therefore higher rate of reaction



Enzymes affecting Rate of Reactions

- Enzymes are biological catalysts which speed up reactions but remain chemically unchanged at the end
- Enzymes function best at optimum temperature and pH level otherwise they may denature and completely stop functioning

6.8. Rates of Reaction Graphs

Interpreting graphs:

A graph with a steeper gradient at the beginning and reaching a horizontal gradient faster depicts a high rate of reaction.

Gas Evolved	Mass Loss	Colour Change
Volume of gas produced (cm³) Time (min)	Mass of system (g) Time (min)	
If a gas evolves, measure volume of gas produced per unit time using a gas syringe	If a gas evolves, measure loss in mass \n per unit time by placing on a balance then putting \n a cotton wool on top to allow gas to pass but not to enter	If a colour change , measure the time taken to turn cloudy

6.9. Reversible Reactions and Equilibrium

The symbol of reversible reactions are $\rightleftharpoons$

- The products can then react with each other or decompose to form the reactant molecules.
- Two equations you should know:
 - $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (blue) $\rightleftharpoons$ CuSO_4 (white) + H_2O
 - $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ (pink) $\rightleftharpoons$ CoCl_2 (blue) + H_2O
- (anhydrous by heating; hydrated form by adding water)

Reversible Reaction in a closed system when Equilibrium:

- Rate of forward reaction = rate of reverse reaction

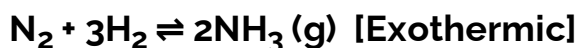
- Concentrations of all reactants and products remain constant, and are no longer changing

Equilibrium

- Le Châtelier's Principle:** if conditions of an equilibrium are changed, the position of equilibrium moves to oppose change
- Temperature:** Temperature lowered; equilibrium moves in exothermic direction. Temperature increases; equilibrium moves in endothermic direction.
- Pressure:** Pressure raised; equilibrium moves to side with fewest gas molecules. Pressure lowered; equilibrium moves to side with most gas molecules.
- Concentration:** Decreasing reactant concentration or increasing product concentration; equilibrium moves to reactant side. Increasing reactant concentration or decreasing product concentration; equilibrium moves to product side.

Oxidising Agents are **Electron Acceptors** and Reducing Agents are **Electron Donor**.

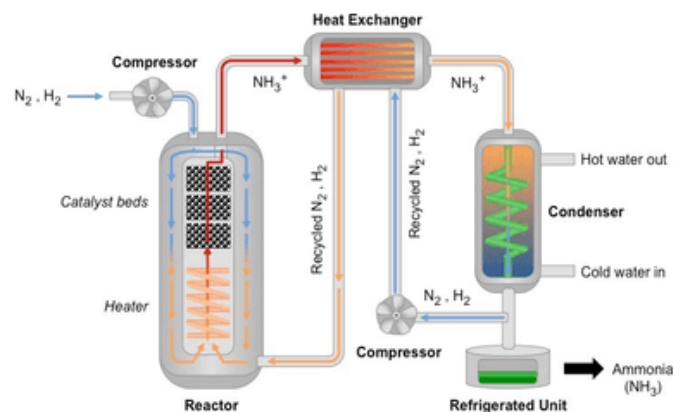
6.10. Haber Process



Materials (Reactants) in Haber Process

Nitrogen: From the air

Hydrogen: From the reaction between Methane with Steam



Conditions

Temperature: 450°C

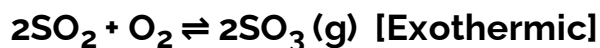
Pressure: 200 atm (20,000 kPa)

Catalyst: Iron catalyst

Why are these conditions chosen in the Haber Process?

- Pressure is not too high** - (Although increase pressure, is more yield, however more pressure will require more expensive equipment and it can explode).
- Optimum temperature** - (If lower temperature, more ammonia yield, however too low, the rate is uneconomical).

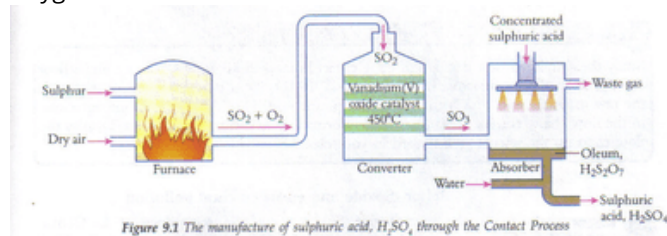
6.11. Contact Process



Materials (Reactants) in Contact Process

Sulfur Dioxide: Burning sulfur or Roasting Sulfide ores

Oxygen: From the air



Conditions

Temperature: 450°C

Pressure: 2 atm (200 kPa)

Catalyst: Vanadium (V) oxide catalyst

Why are these conditions chosen in the Contact Process?

1. **Pressure is high** - (Increase pressure, more yield)
2. **Optimum temperature** - (If lower temperature, more sulfuric acid yield, however too low, the rate is uneconomical.)

6.12. Redox

- A simultaneous oxidation and reduction reaction

Oxidation	Reduction
Loss of electrons	Gain of electrons
Gain of oxygen	Loss of oxygen
Loss of hydrogen	Gain of hydrogen

Note: **Roman numerals** next to an element's name are the **oxidation number of an element in a compound**, eg. Iron (II) and Iron (III). The reaction product formula depends on this.

Reducing agents are oxidized and oxidizing agents are reduced

Identifying redox reactions

1. Oxidation state:

Oxidation state highlights electron movement in a reaction
eg. $\text{CuO} + \text{Mg} \rightarrow \text{MgO} + \text{Cu}$

$\text{Cu}^{2+} + \text{Mg} \rightarrow \text{Mg}^{2+} + \text{Cu}$ [oxide is a spectator and is removed as it doesn't change its oxidation state]

Here copper (II) is reduced while magnesium is oxidized; the reaction is redox

2. Indicators

- Acidified Aqueous Potassium manganate (VII)** is a deep purple oxidising agent; when added to a reducing agent changes from purple to colourless

- Aqueous Potassium iodide** is a reducing agent; when added to an oxidising agent changes colorless to yellow-brown

Redox Reactions by changes in Oxidation Number

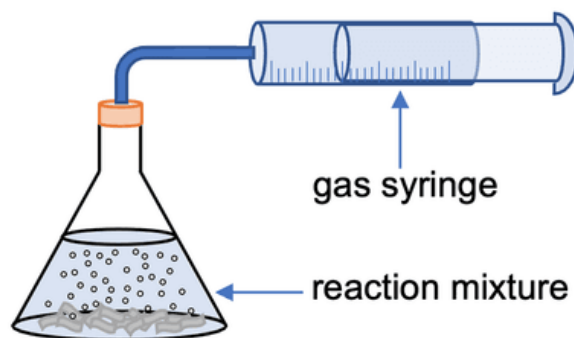
Rules	Example
The oxidation numbers of element in their uncombined state is zero	In H_2 , the oxidation number of H is "0"
The oxidation number of a monatomic ion is the same as the charge of the ion	In Zn^{2+} , the oxidation number is "+2"
The sum of the oxidation numbers in a compound is zero	In H_2O , the oxidation number of H is +1 and O is -2, $(2 \times +1) + (1 \times -2) = 0$
The sum of the oxidation numbers in an ion is equal to the charge of the ion	In MnO_4^- , the oxidation number of Mn is +7, therefore $(1 \times -7) + (4 \times -2) = -1$

6.13. Evaluating Practical Methods

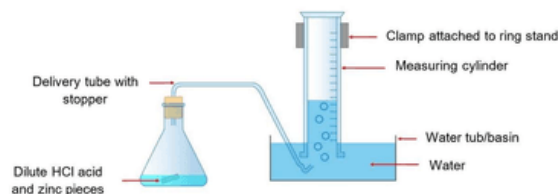
1. Change in mass method

- Not suitable for experiment where hydrogen gas is given off (because density decreases, too small mass)

2. Easier to set up **gas syringe** and obtain reliable measurements



3. **Inverted Measuring cylinder** to collect gas over water, is harder to set up



7. Acids, Bases and Salts

7.1. Properties of Acids

An acid is a **substance that produces hydrogen ions** (H^+) when dissolved in water.

Definition of Acids: **Proton donors.**

Indicators

- Have pH between 1 (strong) and 6 (weak)
- Turns blue **litmus** red
- Turns **methyl orange** indicator red
- Colourless in **Thymolphthalein**

Weak and Strong Acids

1. **Strong acids:** completely dissociated in aqueous solution producing lots of H^+ ions

e.g: $\text{HCl (aq)} \rightarrow \text{H}^+ \text{ (aq)} + \text{Cl}^- \text{ (aq)}$

2. **Weak acids:** partially dissociated in aqueous solution producing few H^+ ions

e.g: $\text{CH}_3\text{COOH (aq)} \rightleftharpoons \text{H}^+ \text{ (aq)} + \text{CH}_3\text{COO}^- \text{ (aq)}$

Chemical properties

- Acid + metal $\rightarrow$ salt + hydrogen gas
- Acid + base $\rightarrow$ salt + water
- Acid + metal carbonate $\rightarrow$ salt + carbon dioxide + water

7.2. Properties of Bases

Bases are substances which neutralize acids to form a salt and water only.

Definition of Bases: They are **proton acceptors** (form OH^- ions)

Indicators

- Have pH between 8 (weak) and 14 (strong)
- Turns red **litmus** blue
- Turns **methyl orange** indicator yellow
- Turns Blue in **Thymolphthalein**

Weak and Strong Alkalis (soluble bases)

1. **Strong alkalis:** completely dissociates in aqueous solution producing lots of OH^- ions

e.g: $\text{NaOH (aq)} \rightarrow \text{Na}^+ \text{ (aq)} + \text{OH}^- \text{ (aq)}$

2. **Weak alkalis** partially ionize in water producing OH^- ions

e.g: $\text{NH}_4\text{OH (aq)} \rightleftharpoons \text{NH}_4^+ \text{ (aq)} + \text{OH}^- \text{ (aq)}$

Chemical properties

- Base + acid $\rightarrow$ salt + water (+ CO_2 when base is a metal carbonate)
- Base + ammonium salt $\rightarrow$ salt + ammonia gas + water

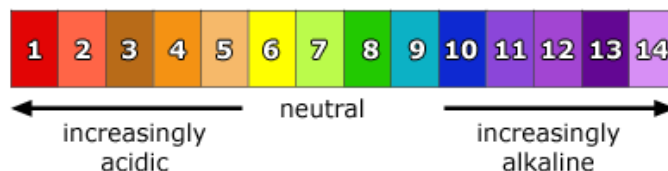
7.3. Neutral

- Neutral substances are pH 7.

- pH scale

pH is the concentration of H^+ ions per dm^3 of solution

Universal indicator solution is used to determine the pH of a substance by matching the color change to the pH color chart.



7.4. Types of Oxides

- Metal oxides are **basic** in nature e.g. Copper oxide and Calcium oxide
- Non-metal oxides are **acidic** in nature e.g. sulfur dioxide and carbon dioxide
- Aluminum, zinc form **amphoteric oxides** e.g. zinc oxide
- Oxides that react with neither acids nor bases are **neutral** e.g. nitrous monoxide and carbon monoxide

Colours of Transition Metal Compound

Metal Compounds	Colour
Copper (II) Sulfate	Blue
Copper (II) Oxide	Black
Copper (II) Carbonate	Green
Manganese (IV) Oxide	Black
Iron (II) Salts	Pale Green
Iron (III) Salts	Brown or Orange

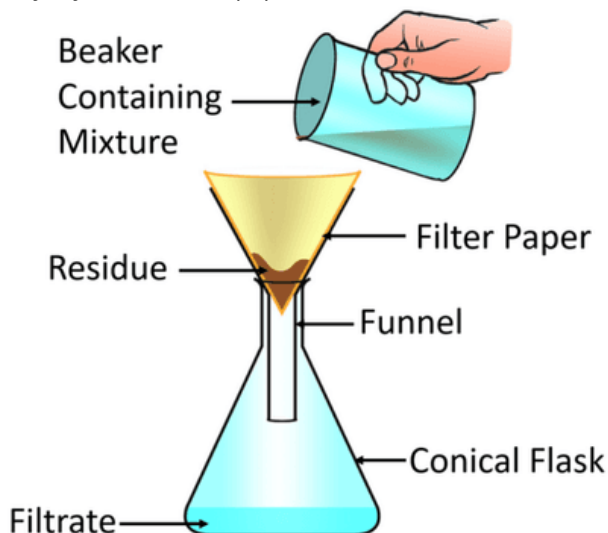
7.5. Preparation of Soluble & Insoluble Salts

Soluble Salts

Method A: Soluble Salts from Excess Insoluble Bases (metal, metal oxide, carbonates)

1. Warm acid (increases speed of reaction)
2. Add excess of reactant + stir
3. Filter mixture
4. Transfer to evaporating basin
5. Heat using bunsen burner
6. Leave to cool to crystallisation point
7. Wash crystals with distilled water

8. Dry crystals on filter paper



Method B: Titration

1. Place known volume of alkali using volumetric pipette into a conical flask
2. Add indicator (e.g. thymolphthalein)
3. Titration: add acid using burette until end point has reached
4. Record volume of acid added
5. Repeat without indicator
6. Transfer to evaporating basin
7. Heat with bunsen burner
8. Leave to cool to crystallisation point
9. Wash crystals with distilled water
10. Dry crystals on filter paper

Insoluble Salt

Precipitation (Insoluble Solid forms between two aqueous solutions)

1. Mix two soluble salts
2. Filter to remove the precipitate
3. Wash the precipitate with distilled water
4. Leave to dry

7.6. Preparation of Salts

- A salt is a compound formed when all the hydrogen atoms of an acid are replaced by a metal.
- Naming salts involves 2 parts; the name of the metal and the acid ending
eg. calcium + hydrochloric acid = calcium chloride
- Water of Crystallisation is the **water molecules present in hydrated crystals**.

Type of Salt	Acid used
Sulfate	Sulfuric acid
Nitrate	Nitric acid

Type of Salt	Acid used
Chloride	Hydrochloric acid
Ethanoate	Ethanoic acid

- Salts can either be soluble or insoluble

Soluble Salts	Insoluble Salts
All sodium, potassium and ammonium salts	None
All nitrates	None
Chlorides	Except silver and lead
Sulfates	Except barium, lead and calcium
Potassium, sodium and ammonium carbonates	All other carbonates
Sodium, Potassium and Ammonium Hydroxides (partially calcium hydroxide)	Nearly all hydroxides

Identification of ions and gases

7.7. Test for Aqueous Cations

Cations	with aqueous NaOH	with aqueous Ammonia
Aluminum (Al^{3+})	White soluble precipitate, turns colorless in excess	White precipitate, insoluble in excess
Ammonium (NH_4^+)	Ammonia Gas produced on warming	
Calcium (Ca^{2+})	White precipitate, insoluble in excess	Faint or no precipitate
Copper (Cu^{2+})	Light Blue precipitate, insoluble in excess	Light Blue precipitate, soluble in excess to give a dark blue solution
Iron(II) (Fe^{2+})	Green precipitate, insoluble in excess	Green precipitate, insoluble in excess
Iron(III) (Fe^{3+})	Red-brown precipitate, insoluble in excess	Red-brown precipitate, insoluble in excess
Zinc (Zn^{2+})	White precipitate, soluble and turns colorless in excess	White precipitate, soluble and turns colorless in excess
Chromium (Cr^{3+})	Green precipitate, soluble in excess	Green precipitate, insoluble in excess

7.8. Test for Anions

Sulfate ions (SO_4^{2-}):

Add dilute nitric acid, then add aq. barium nitrate
White precipitate formed

Sulfite ions (SO_3^{2-}):

Add acidified aqueous potassium manganate (VII) and heat
Color changes from purple to colourless

Halide ions:

Add nitric acid, then aqueous silver nitrate

Chloride (Cl^-)	White precipitate
Bromide (Br^-)	Cream precipitate
Iodide (I^-)	Yellow precipitate

Nitrate ions (NO_3^-):

Add aqueous sodium hydroxide then add warm aluminum foil
Pungent gas produced, turns damp red litmus blue

Carbonate ions (CO_3^{2-}):

Add dilute hydrochloric acid

If bubbles/ gas produced turn limewater cloudy, carbonate ion present

7.9. Test for Gases

Gas	Test and test result
Ammonia (NH_3)	Damp red litmus paper turns blue
Carbon dioxide (CO_2)	Turns Limewater milky
Chlorine (Cl_2)	Bleaches red/blue litmus paper
Hydrogen (H_2)	Place lighted splint, squeaky pop
Oxygen (O_2)	Place glowing splint, splint relights
Sulfur Dioxide (SO_2)	Turns Acidified Aqueous Potassium Manganate (VII) from purple to colourless

7.10. Flame Tests

Metal Ion	Flame Colour
Lithium	Red
Sodium	Yellow
Calcium	Orange-red
Potassium	Lilac
Barium	Light-Green
Copper (II)	Blue-Green

8. The Periodic Table

- The Periodic table is a method of classifying elements.
- Elements are arranged in order of increasing atomic number (each proceeding element has one more proton)

- Made up of rows called periods and columns called groups; the position of an element helps determine its electronic configuration
- Period number: number of electron shells
- Group number: number of outer shell electrons
- Elements in the same group have similar chemical properties.

The Periodic Table of Elements

Key:
atomic symbol
atomic number

lanthanoids
actinoids

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

8.2. Periodic Trends

- Table moves from metals on the left to non-metals on the right.
- Down a group of metals, elements become more reactive
- With non-metals, going down a group, reactivity decreases

8.3. Group I - Alkali Metals

- Group I metals: Lithium, sodium and potassium

Chemical Properties	Physical Properties
Readily react with oxygen and water; stored in oil	Good conductors of heat and electricity
React violently with chlorine	Soft and easy to cut
Burst into flames when heated with oxygen [red flame for lithium; yellow flame for sodium; lilac flame for potassium]	Shiny when freshly cut
Produce soluble white compounds.	Melting points decrease down the group.
React with water to form alkaline metal and hydrogen gas	Increase density down the group

- Predicting the properties of other Group I alkali metals: Rubidium, Caesium and Francium [reactivity increases down the group]

Element	Reaction with water
---------	---------------------

Element	Reaction with water
Lithium	Floats and gives off hydrogen gas (effervescence)
Sodium	Vigorous Reaction and moves very quickly
Potassium	Explosive Reaction, lilac flame

8.4. Transition Metals

Physical Properties

- High melting & boiling points
- Malleable and ductile
- Good conductors of heat & electricity
- High density

Chemical Properties

- Act as catalysts
- Form colored compounds
- Variable Oxidation Numbers (Iron (II) or Iron (III))

8.5. Group VII - Halogens

Properties	Patterns
States and Colors, at RTP: Fluorine- Yellow-green gas Chlorine- Pale Yellow-Green gas Bromine- Red-Brown liquid Iodine- Grey-Black solid	Down the group; size, mass and density increase
Poisonous	Down the group, color darkens
Diatomic; form halide ions in displacement reaction	Reactivity decreases down the group, because it has to gain an electron, so the closer the electron is to the positive nucleus the more easily it will be gained, so atoms with fewer shells will react more easily.
Do not conduct electricity	melting point increases down the group

8.6. Group VIII - Noble Gases

Properties	Uses
Density increases down the group	Helium- filling balloons
Monoatomic and colourless	Argon – Lamps
M.P. and B.P. increases down the group	Neon – advertising signs
Don't conduct electricity	
Inert & stable, due to full outer shell electrons	

9. Metals

9.1. Properties of Metals

Physical Properties

1. Good conductors of heat (when molten) and electricity
2. High melting and boiling points
3. Malleable and Ductile
4. High densities
5. Solids at room temperature (except mercury in liquid)

Chemical Properties

1. Metals + Acids → Salt + Hydrogen
2. Metal + Oxygen → Metal Oxide
3. Metal + Water → Metal Hydroxide + Hydrogen
4. Metal + Steam → Metal Oxide + Hydrogen

Non-Metals

Properties of Non Metals

Physical Properties

1. Poor thermal conductor
2. Poor conductor of electricity (Except graphite)
3. Brittle and not malleable
4. Solids and gases at room temperature (bromine is liquid)

9.2. Uses of Metal

- **Aluminum**
 - Manufacture of Aircraft/Cars (Low density)
 - Food Containers (Resistant to corrosion)
 - Overhead electrical cable (Good conductor of electricity/ductile)
- **Zinc**
 - Galvanizes Iron = coats it to stop it rusting (protective coating)
 - Alloys – brass/bronze
 - Batteries
 - Sacrificial Protection
- **Copper**
 - Electrical Wiring (Good conductor of electricity/Ductile)

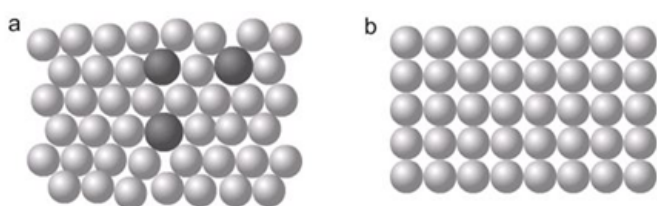
9.3. Alloys

- An alloy is a mixture of two or more metals, or a metal and non-metal
- Alloys are useful because they are harder and stronger/useful than pure metal, it can also resist rusting

with air and water.

Alloy	Made from	Special Properties	Uses
Brass	Copper and zinc	Stronger and more resistant to corrosion/rusting	Electrical fittings, car radiators
Stainless steel	Iron, carbon, chromium and nickel	resistant to corrosion/rusting	Kitchen sinks, cutlery, surgical instruments

- Metal alloys, due to the irregularity in atom sizes and structure which stop layers from sliding over each other, are stronger
- This is what the structure of an alloy (a) looks like, compared to a pure metal (b).



9.4. Reactivity Series

Note: Aluminum- despite its high placement in the reactivity series- is seemingly unreactive due to its protective aluminum oxide layer

General Reactivity series (descending order of reactivity)	Chemical reactivity of metals
K - Potassium Na - Sodium Ca - Calcium Mg - Magnesium Al - Aluminum C - Carbon Zn - Zinc Fe - Iron Pb - Lead H - Hydrogen Cu - Copper Ag - Silver Au - Gold	Water: 1. metal + cold water → metal hydroxide + hydrogen 2. metal + steam → metal oxide + hydrogen
Everything above hydrogen can displace hydrogen from its acid, and hydrogen cannot reduce their oxides.	Oxygen: metal + oxygen → metal oxide

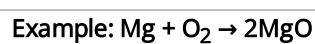
General Reactivity series (descending order of reactivity)	Chemical reactivity of metals
Metals above carbon, their oxides cannot be reduced by carbon	Dilute acids: In a metal and acid reaction, the hydrogen atom in the acid is replaced by the metal atom to form a product of salt and hydrogen

Copper, Silver and Gold cannot react with Dilute HCL, because its **too unreactive**

9.5. Displacement Reactions

In these reactions, metals compete for oxygen or anions

- Oxidisation is the loss of electrons
- Reduction is the gain of electrons
- The more reactive metal will displace the less reactive metal from oxygen or an anion.
- If more reactive metal has oxygen or an anion, no reaction occurs
- The bigger the difference in reactivity between the two metals, the faster the reaction



Here Magnesium is oxidized while the Oxygen atom is reduced

9.6. Corrosion in metals

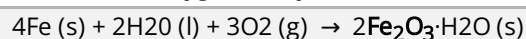
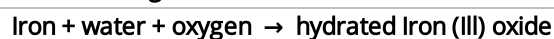
Rusting is the corrosion of iron and steel to form rust (hydrated iron (III) oxide) via oxidation



The conditions for rusting to occur:

- Water
- Air containing Oxygen

Equation for Rusting of Iron



Two ways of preventing rust:

1. Barrier Method

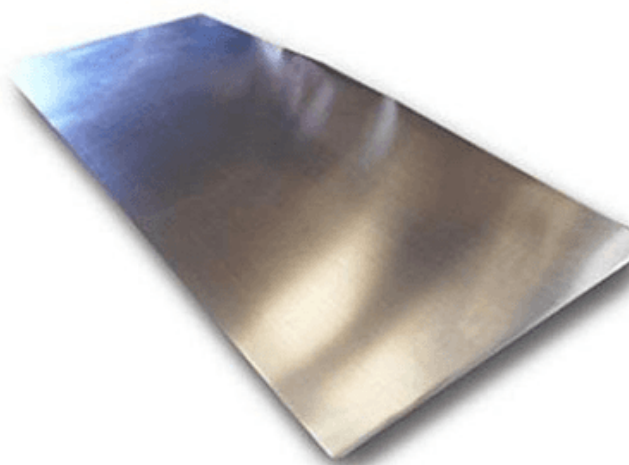
- Coating a material to prevent the iron/steel from being in contact with water and oxygen (Painting, Greasing & Plastic Coating)

2. Sacrificial Method

- When more reactive metals corrode in preference to less reactive metals where they lose electrons in preference to iron.

Galvanisation of Iron

It is the **protection of iron and steel objects** by coating with a **layer of zinc**



Barrier method due to the zinc layer preventing exposure to air and water and **Sacrificial protection** due to zinc being more reactive than iron and therefore corroding in preference to iron.

9.7. Extraction of Metals

- Process of separating a particular metal from its compound; metal ore
- Ores more difficult to decompose from gold to potassium (expensive)

Metal	Extraction Method
K - Potassium Na - Sodium Ca - Calcium Mg - Magnesium Al - Aluminum	Reduction via electrolysis
Carbon	
Zn - Zinc Fe - Iron Pb - Lead	Reducing by Carbon
Hydrogen	

Metal	Extraction Method
Cu - Copper Ag - Silver Au - Gold	Occur naturally

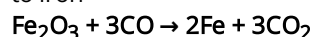
9.8. Extraction of Iron

- Ore haematite (Fe_2O_3) + Impurities
- Burning of coke (Carbon) to **provide heat and produce carbon dioxide**

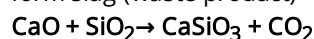
- Ore is crushed and mixed with carbon and limestone (CaCO_3) and transferred into the blast furnace

$\text{C} + \text{O}_2 \rightarrow \text{CO}_2$ (exothermic)
$\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ (thermal decomposition)
$\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$

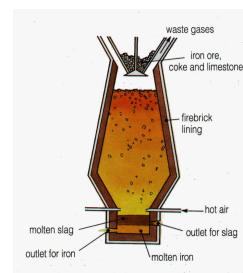
- Carbon monoxide reduces Iron(III) oxide from the ore to iron



- The calcium oxide reacts with impurities like silica to form slag (waste product)



- Uses of slag include making roads and cement



The remaining waste gases that leaves the blast furnace are **Carbon dioxide, carbon monoxide, nitrogen gas.**

10. Water

10.1. Tests for Water

Test	Type of test	Positive result
Anhydrous Cobalt (II) Chloride	Chemical	Turns from blue to pink
Anhydrous Copper (II) sulfate	Chemical	Turns from white to blue
Test melting and boiling point	Physical	M.P at 0°C and B.P at 100°C

Distilled Water is used in practical chemistry rather than tap water because it has fewer chemical impurities.

10.2. Natural Sources of Water

Water is an important source in the natural world. However with the changing world and massive urbanisation. **Our water may contain substances such as:**

1. Dissolved Oxygen (**this is important for aquatic life**)
2. Metal Compounds (Provides **essential minerals for life** - **however some are toxic**)
3. Plastics (**harm aquatic life**)
4. Sewage (**contains harmful microbes which cause diseases**)
5. Harmful microbes
6. Nitrates from fertilizers
7. Phosphate from fertilizers and detergents (**lead to deoxygenation of water and damage to aquatic life (Eutrophication)**)



10.3. Treatment of Domestic Water Supply

1. Water is pumped into screens to **remove solid insoluble impurities**
2. A **sedimentation** process making small clay pieces stick together and are then removed
3. The water is then undergoes **filtration** through layers of sand and gravel to **remove larger, insoluble debris**.
4. **Carbon** is also added into filtered water to **remove tastes and odour in the water**.
5. The **chlorination process** adds chlorine gas is bubbled into the water to **kill bacteria and other microbes**; the acidic effect on the water is reversed by adding an

alkali, sodium hydroxide



Uses of Water

Home	Industry
Drinking, cooking and washing	Water jet cutting and water blasting
In car radiators, for gardens and plants	As a solvent in refining ores
	Generating hydroelectricity

10.4. Fertilisers

Fertilisers are substances added to the soil and taken up by plants to increase the crop yield. Substances which contains inside Fertilisers are Ammonium Salts and Nitrates.



NPK - Nitrogen, Phosphorus, Potassium Fertilisers, and these are found inside fertilisers, which are essential to **provide improved growth for plants**.

Function of Elements

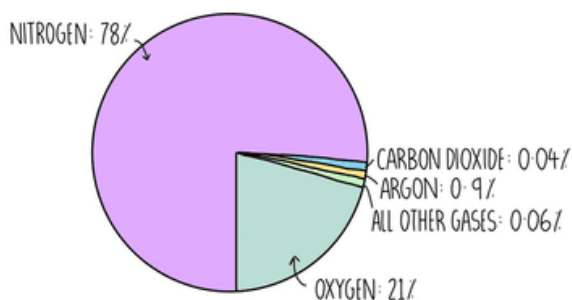
1. Nitrogen - Makes chlorophyll and protein. Promotes healthy leaves
2. Phosphorus - Promotes healthy roots
3. Potassium - Promotes growth and healthy fruits and flowers

Reaction with any alkali substance (except ammonia) displaces ammonia from its compound, for example:
Calcium hydroxide + ammonium chloride → Calcium chloride + Ammonia + Water

10.5. Air Quality and Climate

Components of clean air:

- Primary- **Nitrogen (78%), Oxygen (21%)**
- Secondary- **Noble gases (mainly Argon) and carbon dioxide (1%)**

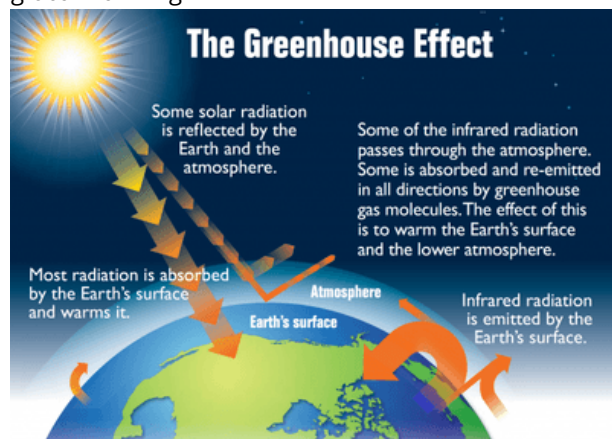


10.6. Pollutants in Air

Pollutant	Source	Negative impact
Carbon monoxide (CO)	Incomplete combustion of carbon-containing fuels (ex. Internal combustion engines)	Binds with haemoglobin, constricting oxygen supply in cells; leads to fatigue/ death
Carbon Dioxide (CO ₂)	Complete Combustion of Carbon Containing Fuels	Increased global warming leading to climate change.
Methane (CH ₄)	Decomposition of vegetation and waste gases from digestion in animals	Increased global warming leading to climate change.
Sulfur Dioxide (SO ₂)	Combustion of Fossil Fuels which contain sulfur compounds	Causes acid rain.
Nitrogen Oxides (NO _x)	High temperatures that trigger a reaction between N ₂ and O ₂ (from air)	Causes respiratory problems, photochemical smog; contributes to acid rain
Lead Compounds	Combustion of leaded fuels	Damages brain and nerve cells in young children

Greenhouse Gases Impact to Global Warming

1. **Short wavelength radiation** from Sun reaches the Earth's surface
2. Some thermal energy is **absorbed and heats oceans/lands**
3. Some thermal energy is **radiated by Earth as longer wavelength radiation**
4. Greenhouse gases absorb some of the infrared radiation and **re-emit in all directions**
5. Some infrared radiation comes back to Earth's surface and this reduces the heat loss to space and leads to global warming



To reduce the effect of climate change:

- Planting trees
- Reduction in livestock farming
- Less use of fossil fuels
- Increased use of hydrogen and renewable energy (e.g. wind, solar)

To reduce the effect of acid rain occurring:

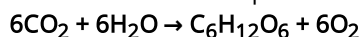
- Reduced emissions of sulfur dioxide by using low-sulfur fuels
- **Flue-gas desulfurisation** with calcium oxide (removing sulfur dioxide from plastic combustion)
- Use of catalytic converters in vehicles

10.7. Photosynthesis

Photosynthesis is the reaction between carbon dioxide and water to produce glucose and oxygen in the presence of chlorophyll and using energy from light.

Word Equation: **Carbon Dioxide + Water → Glucose + Oxygen**

Balanced Chemical Equation:



10.8. Catalytic Converters

1. Present in car exhausts; contains transition metal catalysts of platinum and rhodium
2. Aids redox reactions to neutralize toxic pollutants formed as a result of incomplete fuel combustion (a) Carbon monoxide (b) Nitrogen oxides (c) Unburned hydrocarbons

3. Reaction equations: (a) $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$
 (b) $2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$
 (c) $\text{C}_8\text{H}_{18} + 12\frac{1}{2}\text{O}_2 \rightarrow 8\text{CO}_2 + 9\text{H}_2\text{O}$



11. Organic Chemistry

11.1. Organic Formulae, Functional Group and Terminology

Suffix	Compound type
-ane	Alkane
-ene	Alkene
-anol	Alcohol
-anoic acid	Carboxylic acid
-yl /-anoate	Ester

Homologous series are a group of organic compounds that have similar chemical properties due to being part of the:

- Same Functional Group
- Same General Formula
- Differing from One member to the next by a CH_2 unit
- Displaying a trend in physical properties
- Sharing a similar chemical properties

Essential Terminologies

1. Functional Group is an **atom or group of atoms** that **determine the chemical properties** of a homologous series
2. Structural Isomers are compounds which have the **same molecular formula** but **different structural formula**.
3. Saturated Compounds - Has molecules which all carbon-carbon bonds are **single bonds**
4. Unsaturated Compounds - Has molecules in which **one or more carbon-carbon bonds** are not single bonds.

11.2. Fuels

Common fossil fuels include: coal, natural gas [main constituent: methane] and petroleum

Petroleum is a mixture of hydrocarbons (**Carbon and Hydrogen ONLY**) which can be separated into useful fractions by fractional distillation

Going down the fractions

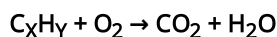
- Increasing chain length
- Increasing temperature
- Lower volatility
- Increase Boiling Points
- Increase Viscosity (harder to flow)

Petroleum fraction	Use
Refinery gas	heating and cooking
Gasoline fraction	fuels for cars
Naphtha fraction	chemical feedstock
Kerosene/paraffin fraction	jet fuel
Diesel oil/gas oil fraction	fuel in diesel engines
Fuel oil fraction	fuel in ships and home heating systems
Lubricating fraction	lubricants, waxes and polishes
Bitumen	making roads

Two types of Combustion

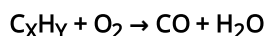
Complete Combustion

A complete combustion reaction with any organic compounds will produce **Carbon Dioxide and Water** as its product. The general equation is:



Incomplete Combustion

An incomplete combustion reaction with any organic compounds will produce **Carbon Monoxide and Water** as its product. The general equation is:



11.3. Alkanes

Alkanes are saturated hydrocarbons [single carbon bonds] that are generally unreactive however they do undergo combustion reactions

General formula = $\text{C}_n\text{H}_{2n+2}$

Methane: CH_4 (n=1)	Ethane: C_2H_6 (n=2)
<pre> H H - C - H H </pre>	<pre> H H H - C - C - H H H </pre>
Propane: C_3H_8 (n=3)	Butane: C_4H_{10} (n=4)
<pre> H H H H - C - C - C - H H H H </pre>	<pre> H H H H H - C - C - C - C - H H H H H </pre>

Substitution reaction [Photochemical Reaction]

Alkanes go through substitutional reactions, where the hydrogen atom is replaced by the atom of another element, under the presence of UV light

- The presence of UV light is to **provide activation energy**

$\text{CH}_4 + \text{Cl}_2 \xrightarrow{\text{light}} \text{HCl} + \text{CH}_3\text{Cl} / \text{CH}_2\text{Cl}_2 / \text{CHCl}_3 / \text{CCl}_4$
Compounds = chloromethane / di/tri/tetrachloromethane

11.4. Alkenes

Alkenes are unsaturated hydrocarbons [at least 1 double bond between 2 carbon atoms]

Have isomers: same molecular formula but different structural formula (placement of double bond shifts) eg. but-1ene and but-2ene

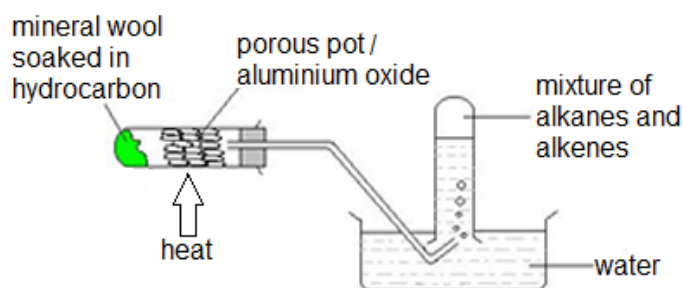
General formula = C_nH_{2n}

Functional group: $\text{C}=\text{C}$ bond

(n=1)	Ethene: C_2H_4 (n=2)
N/A	$\begin{array}{c} \text{H} & & \text{H} \\ & \diagdown & / \\ & \text{C}=\text{C} \\ & / & \diagdown \\ \text{H} & & \text{H} \end{array}$
Propene: C_3H_6 (n=3)	But-1-ene: C_4H_8 (n=4)
$\begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C}-\text{C}=\text{C} \\ & & \\ \text{H} & & \text{H} \end{array}$	$\begin{array}{c} \text{H} & \text{H} & \text{H} & \text{H} \\ & \diagdown & / & & \\ & \text{C}=\text{C} & -\text{C} & -\text{C} & -\text{H} \\ & / & & & \\ \text{H} & & \text{H} & \text{H} & \text{H} \end{array}$

Catalytic Cracking:

- Thermal decomposition reaction, in which an alkane (and sometimes hydrogen) are produced from an alkane.
- Hydrocarbon heated and vapors passed over catalyst (alumina or silica)
- Cracking** always produces short chain compound with a $\text{C}=\text{C}$ bond (match the supply of fractions with demand and produces alkene for feedstock.)
e.g. Cracking of ethane will give ethene and hydrogen



- Butane $\rightarrow$ Ethane + Ethene ; $\text{C}_4\text{H}_{10} \rightarrow \text{C}_2\text{H}_6 + \text{C}_2\text{H}_4$

Distinguish between saturated and unsaturated hydrocarbons

Using Bromine water (orange-brown):

- Saturated: remains orange (unreactive)
- Unsaturated: turns colourless/decolourises

11.5. Alkenes' Addition Reactions

Alkene addition reactions **only formed one product!**

- With bromine:** (the test for saturation)
e.g. ethene (g) + bromine (aq) $\rightarrow$ 1,2-dibromoethane (l)
- With steam [hydration]:** forms alcohols with heat (300°C), pressure (60atm/6000kPa) and an acid catalyst (phosphoric acid)
e.g. ethene (g) + steam (g) $\rightleftharpoons$ ethanol (l)
- With hydrogen [hydrogenation]:** double bond breaks down to form an alkane with heat, pressure (60 atm) and a catalyst (nickel)
e.g. ethene (g) + hydrogen (g) $\rightarrow$ ethane (g)

11.6. Alcohols

General formula = $\text{C}_n\text{H}_{2n+1}\text{OH}$

Functional group: OH

Methanol: CH_3OH (n=1)	Ethanol: $\text{C}_2\text{H}_5\text{OH}$ (n=2)
$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{O}-\text{H} \\ \\ \text{H} \end{array}$	$\begin{array}{c} \text{H} & \text{H} \\ & \\ \text{H}-\text{C} & -\text{C}-\text{O}-\text{H} \\ & \\ \text{H} & \text{H} \end{array}$
Propanol: $\text{C}_3\text{H}_7\text{OH}$ (n=3)	Butanol: $\text{C}_4\text{H}_9\text{OH}$ (n=4)
$\begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C} & -\text{C} & -\text{C}-\text{O}-\text{H} \\ & & \\ \text{H} & \text{H} & \text{H} \end{array}$	$\begin{array}{c} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{H}-\text{C} & -\text{C} & -\text{C} & -\text{C}-\text{O}-\text{H} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{H} \end{array}$

Methods of production:

Fermentation of Aqueous Glucose (for ethanol)	Catalytic Addition of Steam to Ethene
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Fermentation of Aqueous Glucose (for ethanol)	Catalytic Addition of Steam to Ethene
Yeast added to dissolved glucose. Products: ethanol, carbon dioxide and the Temperature between 25-35 °C for optimal enzyme activity (dies after alcohol concentration increases)	The ethene reacts with steam (reversibly) to form ethanol in the following conditions: 300°C, 60 atm (6000 kPa) Catalyst - phosphoric acid [while low temp. gives better yield, high temp. is used for faster rate of reaction]
Slow reaction, produces dilute solution that requires processing. Can only be produced in batches	Fast reaction, produces pure ethanol. Continuous production (no batches)
Produces greenhouse gas (CO ₂)	No greenhouse pollutants
Uses renewable resources	Uses non-renewable resources (crude oil)

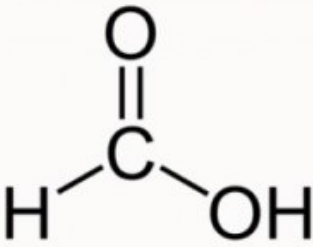
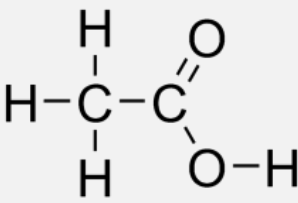
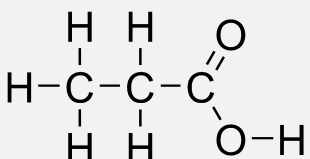
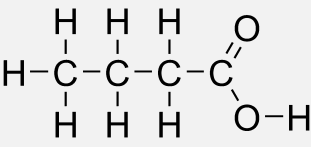
Uses of ethanol:

- Solvent in glues, printing inks & perfumes
- Fuel

11.7. Carboxylic Acids

General formula = C_nH_{2n+1}COOH

Functional group: COOH

Methanoic Acid: CH ₂ O ₂ (n=1)	Ethanoic Acid: CH ₃ COOH (n=2)
	
Propanoic Acid: C ₃ H ₆ O ₂ (n=3)	Butanoic acid: C ₄ H ₈ O ₂ (n=4)
	

Ethanoic acid:

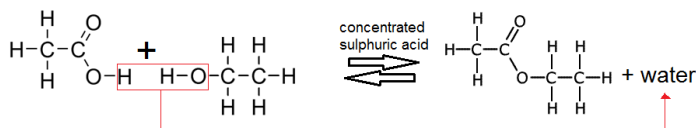
- Weak acid with high pH and low dissociation
- Formed by:
 - Bacterial Oxidation of ethanol (Vinegar Production)
 - With acidified aqueous potassium manganate (VII) (Good Oxidising Agent)

Carboxylic acids react with alcohols (with an acid catalyst/Heat) to give esters, in a condensation reaction, for example:

- Ethanoic acid + ethanol ⇌ ethyl ethanoate + water (alcohol = -yl & carboxylic acid = -oate)
- Carboxylic Acids also have **different structural isomers**.

11.8. Polymers

- Large molecules built up from small units known as monomers.
- Monomers are **a single unit of a substance**



- Polymers can have different linkages depending on the type of polymerization and monomer

Example:

Small units (monomers)	Linkages
Polyester (PET)	Ester
Polyamide & Protein	Amide

Plastics (PET)

Plastics (**polyethylene terephthalate**) are made from polymers, however they are not biodegradable. (Note that: PET can be hydrolysed back to monomers and re-polymerised)



Environment Challenges caused by Plastics

- Disposal in Land fill sites
- Accumulation in Oceans
- Formations of toxic gases from burning

11.9. Properties of Polymers

- Can be moulded under heat and pressure due to its low density
- Low conductivity (retain heat)
- Resistant to corrosion

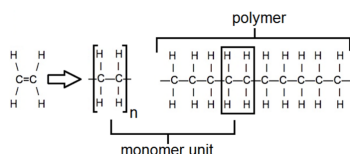
- Non-biodegradable and lead to plastic waste



11.10. Addition and Condensation Polymerization

Addition Polymerization:

- Only occurs in monomers that contain double carbon (C=C) bonds
- Polymers produced using alkene monomers
- Forms only a polymer molecule
- Poly(ethene): is a polymer produced from ethene by addition polymerization

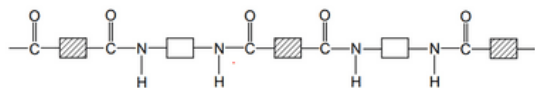


Double bond splits and polymer is formed. *Polymers have no double bonds

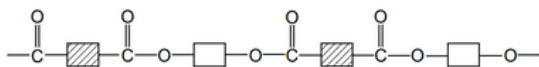
Condensation Polymerization:

- When 2 different monomers are linked together with the removal of a smaller molecule, usually **water** (forms one H₂O molecule per linkage).

- Nylon** [polyamide] is made from a **dicarboxylic acid monomer** and a **diamine monomer** (compound with an NH₂ functional group). Forms **amide linkage**.



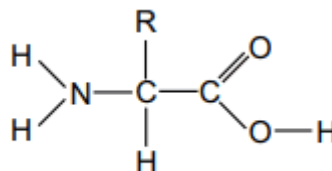
- PET** [polyester] made from a dicarboxylic acid monomer and diols (alcohol with an -OH functional group). Forms **ester linkage**.



11.11. Protein & Natural Polyamides

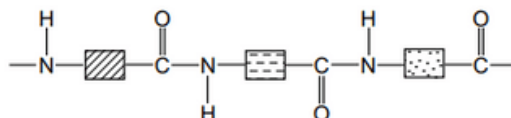
Proteins act as the natural polyamides formed from Amino Acids Monomers

General Structure of Amino Acids:



Proteins:

- Proteins contain the same linkages (amide links) as nylon, but with different units from amino acids monomers. Their main structure you should know is:



12. Experimental Design

12.1. Measurement

	Use	Structure
Poly(ethene)	Plastic bags and gloves, clingfilm (low density), mugs, bowls, chairs, dustbins (high density)	$n \text{ H}_2\text{C}=\text{CH}_2 \rightarrow \left[\text{CH}_2-\text{CH}_2 \right]_n$ ethene poly(ethene) polythene

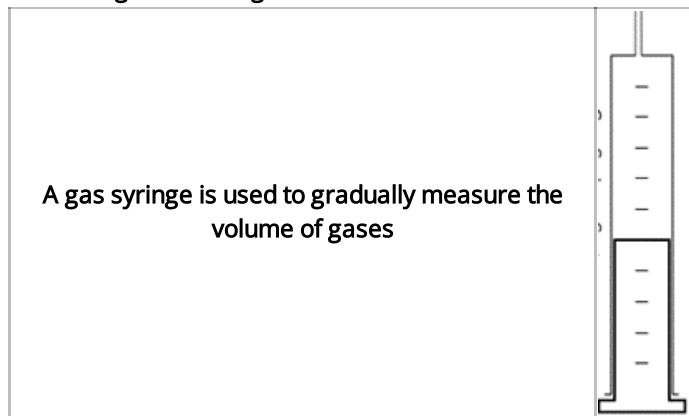
Variable	Unit	Apparatus
Time	min/sec	Stopwatch
Temperature	°C	Thermometer
Mass	grams	Balance

Measuring Volume in liquids:

Approximate measure to most accurate measure

Name	Measuring Cylinder	Volumetric Pipettes [fixed volumes]	Burettes [variable volume]
Image			

Measuring Volume in gases:

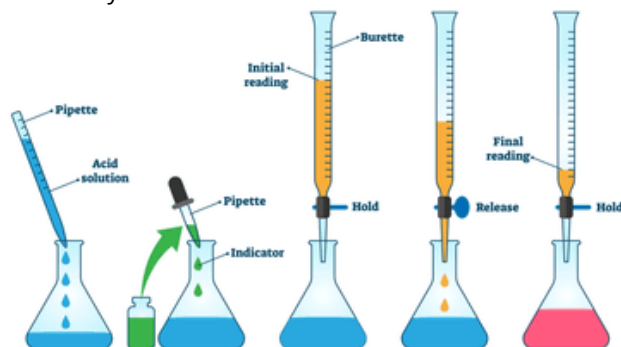


12.2. Mixture of Substances

1. Mixture → A mixture is substance made from two or more other substances not chemically bonded together
2. Solution → A mixture of one or more solutes dissolved in a solvent
3. Solute → A substance that dissolved in a solvent
4. Solvent → A substance that dissolves a solute
5. Saturated Solution → A solution containing the maximum concentration which cannot be dissolved further at a given temperature.

12.3. Acid-Base Titrations

A method of quantitative chemical analysis where an acid is added slowly to a base until it has been neutralised.



In Acid-Base Titrations, we are required to add a few drops of indicator (passes the endpoint) to check whether there is colour changes in the chemical reaction.

Indicators	Colour in Acid	Colour in Neutral	Colour in Base
Thymolphthalein	Colourless	Colourless	Blue
Methyl Orange	Red	Orange	Yellow

12.4. Criteria for Purification

Purity in substances

- Assessing purity

Pure substances	Have a definite, sharp m.p./b.p.
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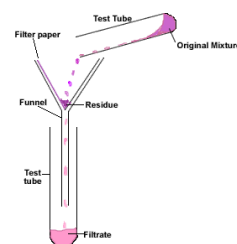
Pure substances	Have a definite, sharp m.p./b.p.
Impure substances	Have a lower m.p and a higher b.p

This assessment of substance purity is important, especially in food consumption, as its intake can be dangerous.

12.5. Filtration

Used to separate an insoluble solid from a liquid

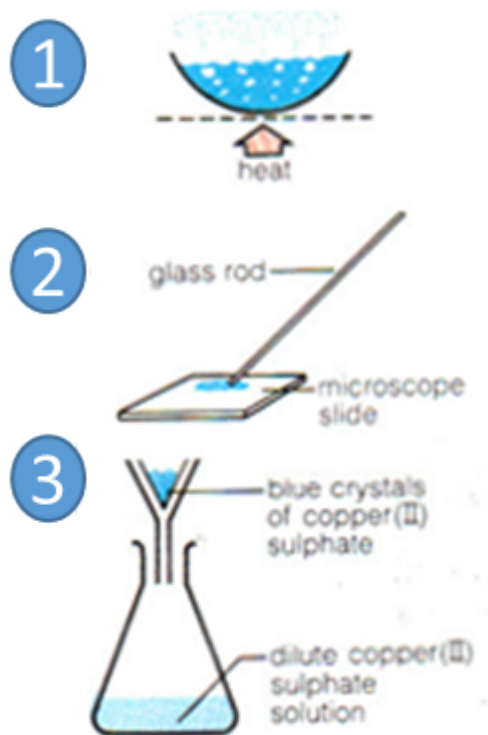
- Mixture goes through a funnel with filter paper, into a flask.
- Insoluble residue remains in the funnel
- Filtrate goes through and collects in flask



12.6. Crystallization

Used to separate dissolved solid from solution

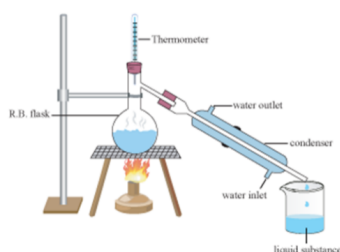
- Solution is heated to increase concentration (solvent evaporates)
- A drop of solution is placed on a slide to check for crystal formation
- Solution is left to cool and crystallise (reaches crystallisation point).
- Crystals are filtered from solution; washed with distilled water and dry between the filter paper



12.7. Simple Distillation

Used to separate a solvent from a solution

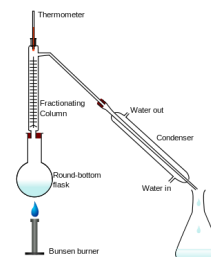
- Impure liquid is heated in a round bottom flask
- When it boils, the steam rises into the attached condenser
- Condenser cools the steams to a pure liquid and it drops into the beaker
- A Thermometer is placed to make sure the **highest boiling point is not exceeded**.



12.8. Fractional Distillation

Used to separate miscible liquids

- Mixture is heated
- Substances, due to their different boiling points, rise in different fractions
- A mixture of gases condense on the beads in the fractional column.
- The beads are heated to the boiling point of the lowest substance, so that substance being removed cannot condense on the beads.
- The other substances continue to condense and will drip back into the flask.
- The beaker can be changed after every fraction



12.9. Separation Methods

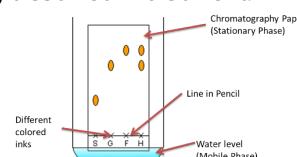
- Can be done by dissolving one in an appropriate solvent
- Then filter one and extract other from solution by evaporation
- If one solid is magnetic, can use a magnet e.g. sand and iron fillings

Component 1	Component 2	Technique	Example
Solid	Solid	Use a solvent	Sand and Salt
Insoluble Solid	Liquid	Use Filtration - Separating the solid and the residue	Copper (II) Oxide and Water
Soluble Solid	Liquid	Crystallization - Liquid is evaporated to leave the solid	Copper (II) sulfate from water
Liquid	Liquid (Miscible)	Distillation (2 Liquids) or Fractional Distillation (More than 2 liquids)	Ethanol & Water (Simple), Crude Oil (Fractional Distillation)

12.10. Chromatography

A Method used to separate substances in a solvent with different solubilities.

1. Drop substance onto the start line (pencil) drawn on chromatography paper
2. Paper is placed in beaker with solvent; the paper must touch the surface of the solvent (**water or ethanol**) while the line must be above the liquid
3. Solvent travels up the paper
4. Different solubilities lead to different travel rates [**high solubility-> high travel rate**]
 - Stationary phase is material on which separation takes place
 - Mobile phase consists of the mixture you want to separate, dissolved in a solvent.



- Interpreting simple chromatograms:

Chromatograms are the visual outputs on the chromatography paper

- **Number of rings/dots = number of substances**
- If two dots travel the same distance up the paper they are the **same substance**.
- Pure substance only gives **one spot**
- Retention Value:

Used to identify a substance, calculated by the formula:

$$\text{Rf Value} = \frac{\text{Distance moved by substance}}{\text{Distance moved by solvent}}$$

Locating Agents

- Used to make colorless chromatograms visible
 - Dry paper in oven
 - Spray it with locating agent
 - Heat it for 10 minutes in oven

CAIE IGCSE Chemistry

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